

AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

Physics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.

Correct Way		Incorrect Ways	
1		1	
		2	
		3	
		4	

Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The total number of significant figures in 0.001007 is

A. 2
B. 4
C. 6
D. 7

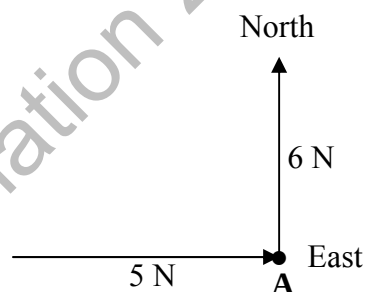
2. In scientific notation, the number 16052017 is expressed as

A. $0.16052017 \times 10^{-1}$
B. 0.16052017×10^1
C. 1.6052017×10^{-7}
D. 1.6052017×10^7

3. In the given figure, a force of 5 N is acting on an object A eastwards and a force of 6 N is acting on it northwards.

The resultant force on the object is equal to

A. $\sqrt{32}$ N
B. 7 N
C. $\sqrt{61}$ N
D. $\sqrt{11}$ N



4. If the dot product of two vectors which are NOT perpendicular to each other is zero, then either one of the vector is

A. a null vector.
B. a unit vector.
C. a positive vector.
D. opposite to the other.

5. The product of the force and the moment arm is called

A. work.
B. torque.
C. energy.
D. equilibrium.

6. A bomb released from a jet plane is an example of

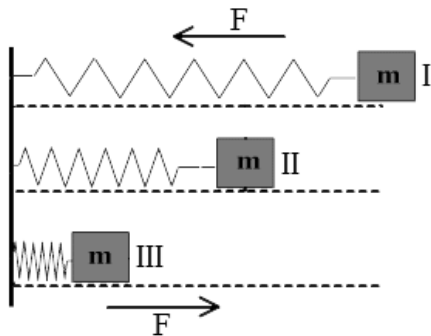
A. linear motion.
B. circular motion.
C. random motion.
D. projectile motion.

7. In a battle field, to obtain the maximum range by a missile fired by an army tank, the missile must be launched at an angle of
- A. 0°
 - B. 30°
 - C. 45°
 - D. 60°
8. Which of the following is one of the sources of non-conventional energy?
- A. Coal
 - B. Bio-mass
 - C. Petroleum
 - D. Water power
9. The energy possessed by a body because of its position is known as
- A. kinetic energy.
 - B. potential energy.
 - C. geothermal energy.
 - D. elastic potential energy.
10. Which of the following statements is TRUE about the term 'work' with reference to physics?
- A. It is a form of energy.
 - B. It is a vector quantity.
 - C. It has the S.I. unit of watt.
 - D. It has the formula, $W = F \times t$.
11. Which of the following statements is TRUE about kinetic energy?
- A. An object in motion possesses kinetic energy.
 - B. An object placed on the ground possesses kinetic energy.
 - C. A light object has more kinetic energy than a heavy object at the same velocity.
 - D. A slow moving object has more kinetic energy than a fast moving object of the same mass.
12. If the angular velocity of clutch plates of a sports car is $12.50 \times 10^{-5} \text{ rad/s}$ and the moment of inertia is $80.04 \times 10^{15} \text{ kg.m}^2$, then the rotational kinetic energy of the clutch plates will be
- A. $8.00 \times 10^9 \text{ J}$
 - B. $4.00 \times 10^8 \text{ J}$
 - C. $1.25 \times 10^9 \text{ J}$
 - D. $6.25 \times 10^8 \text{ J}$

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- The diagram shows a horizontal pipe with four distinct sections labeled A, B, C, and D from left to right. Section A is the widest, section B is narrower than A, section C is the widest of all, and section D is narrower than C but wider than B. Arrows indicate the direction of water flow from left to right. The flow enters section A, passes through B, then C, and finally D.

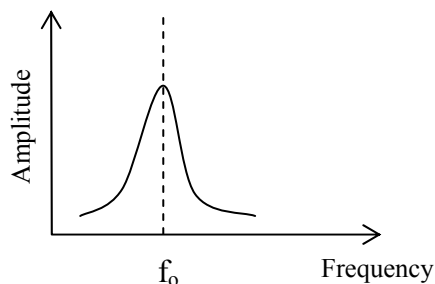
- | A | B | C | D | Order of viscosity |
|-------------|-------------|-------------|-------------|--|
| Tooth paste | Paint | Water | Honey | <div style="position: absolute; top: 0;">Max.</div> <div style="position: absolute; bottom: 0;">Min.</div> |
| Honey | Tooth paste | Paint | Water | |
| Paint | Honey | Tooth paste | Paint | |
| Water | Water | Honey | Tooth paste | |

17. In the given mass-spring system, the potential energy of the block is zero at position/ s

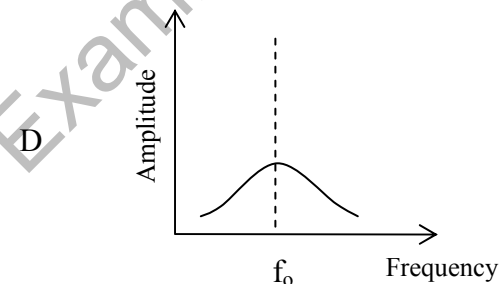
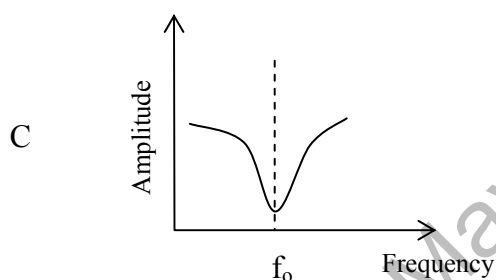
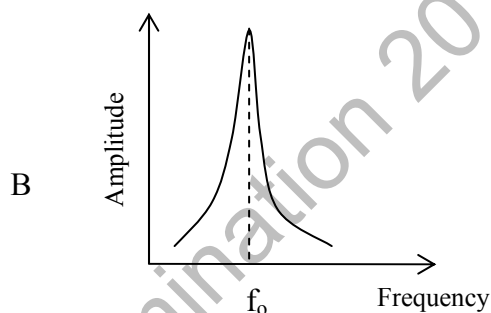
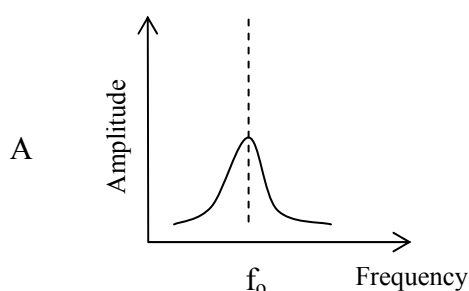


- A. I only
 - B. II only
 - C. I and III
 - D. II and III
18. When a body executes free vibrations without any interference of an external force then the frequency of these free vibrations is known as
- A. forced frequency.
 - B. natural frequency.
 - C. critical frequency.
 - D. original frequency.
19. Which phenomenon occurs when the frequency of free oscillation matches the frequency of forced oscillations?
- A. Beats
 - B. Resonance
 - C. Free oscillation
 - D. Forced oscillation

20.



The given graph shows the response of forced oscillations of a pendulum in air. What will be the response of its forced oscillations in a vacuum?



21. The apparent change in the frequency of sound due to a relative motion between a source and an observer is known as
- Bragg's law.
 - Doppler's effect.
 - Laplace's formula.
 - Superposition of waves.
22. The place where compressions of one wave combines with the rarefactions of another wave to cancel each other is called a
- dead beat.
 - null point.
 - zero point.
 - node point.

23. The narrow slits of a diffraction grating are 5×10^{-4} m apart. If a light ray of 5×10^{-7} m wavelength passes through the narrow slits, then the value of angle observed at the third-order bright spot is
- A. 0.085°
 - B. 0.057°
 - C. 0.028°
 - D. 0.017°
24. All of the following are the conditions necessary for the interference of light EXCEPT
- A. light waves should be monochromatic.
 - B. sources of the light waves must be coherent.
 - C. identical light waves should have constant phase difference.
 - D. size of obstacle must be similar to the wavelength of light waves.
25. Which of the following phenomena reveals that light waves are transverse waves?
- I. Diffraction
 - II. Polarisation
 - III. Interference
- A. I only
 - B. II only
 - C. I and III
 - D. II and III
26. Which of the following relations describes the Boyle's Law?
- A. Pressure inversely proportional to volume
 - B. Volume directly proportional to temperature
 - C. Pressure directly proportional to temperature
 - D. Number of moles inversely proportional to volume
27. If the volume of a thermodynamics system increases, then the work done by the system will be
- A. zero.
 - B. infinite.
 - C. positive.
 - D. negative.
28. In a thermodynamic system, if there is no change in temperature and internal energy, then the process is called an
- A. isobaric process.
 - B. adiabatic process.
 - C. isochoric process.
 - D. isothermal process.

29. In an irreversible process of a thermodynamics system, there is
- A. a loss of heat.
 - B. a decrease in volume.
 - C. an increase in pressure.
 - D. no change in temperature.
30. The efficiency of Carnot Cycle depends upon change in the
- A. mass of active substance.
 - B. volume of active substance.
 - C. pressure on the hot reservoir.
 - D. temperature of hot and cold reservoirs.

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